第四章　静电场中的电介质

§1 电介质对电场的影响

§2 电介质的极化

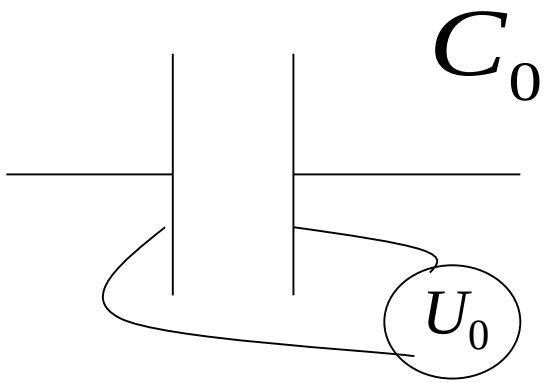
§3 电位移矢量 $\vec{D}$

(有电介质时的高斯定理)

§4 有电介质时静电场的能量

§1 电介质对电场的影响

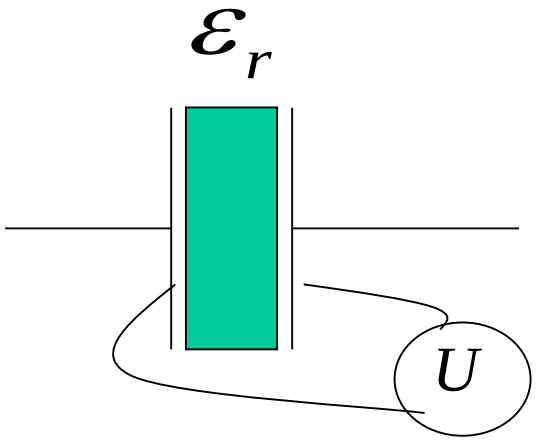
实验现象



$$U = \frac{U_0}{\varepsilon_r}$$

一般

$$\varepsilon_r > 1$$

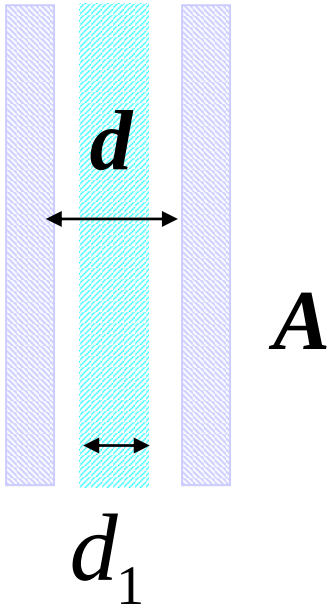


$$\boxed{C = C_0 \varepsilon_r}$$

§2 电介质的极化

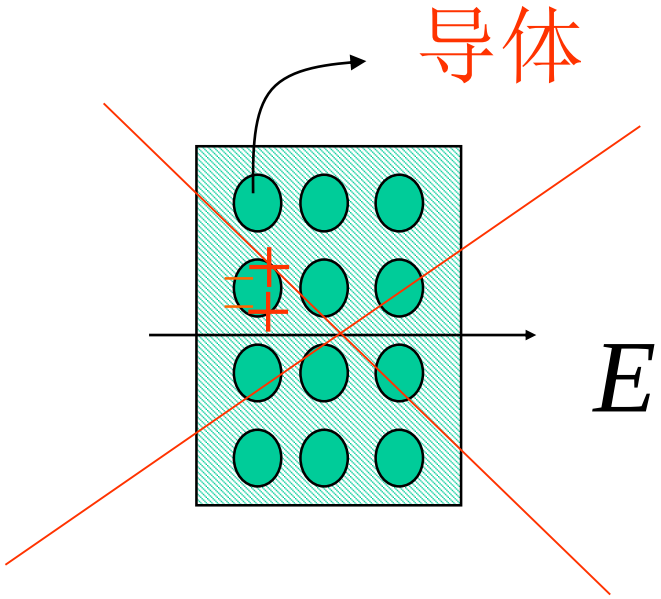
一. 电介质的微观图象

平行板



插入导体板

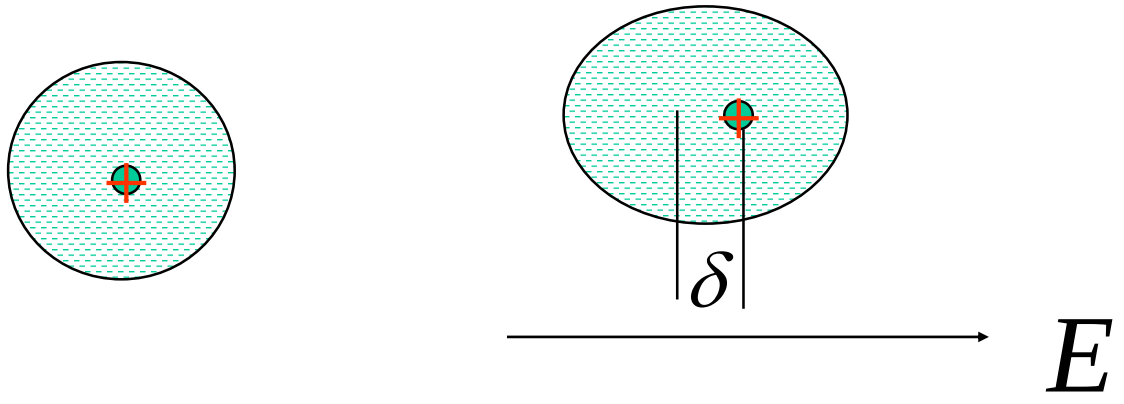
$$U' = E(d - d_1)$$



以为绝缘体（电介质）

对电场反应的是分子或原子

无极分子**non-polar molecules**

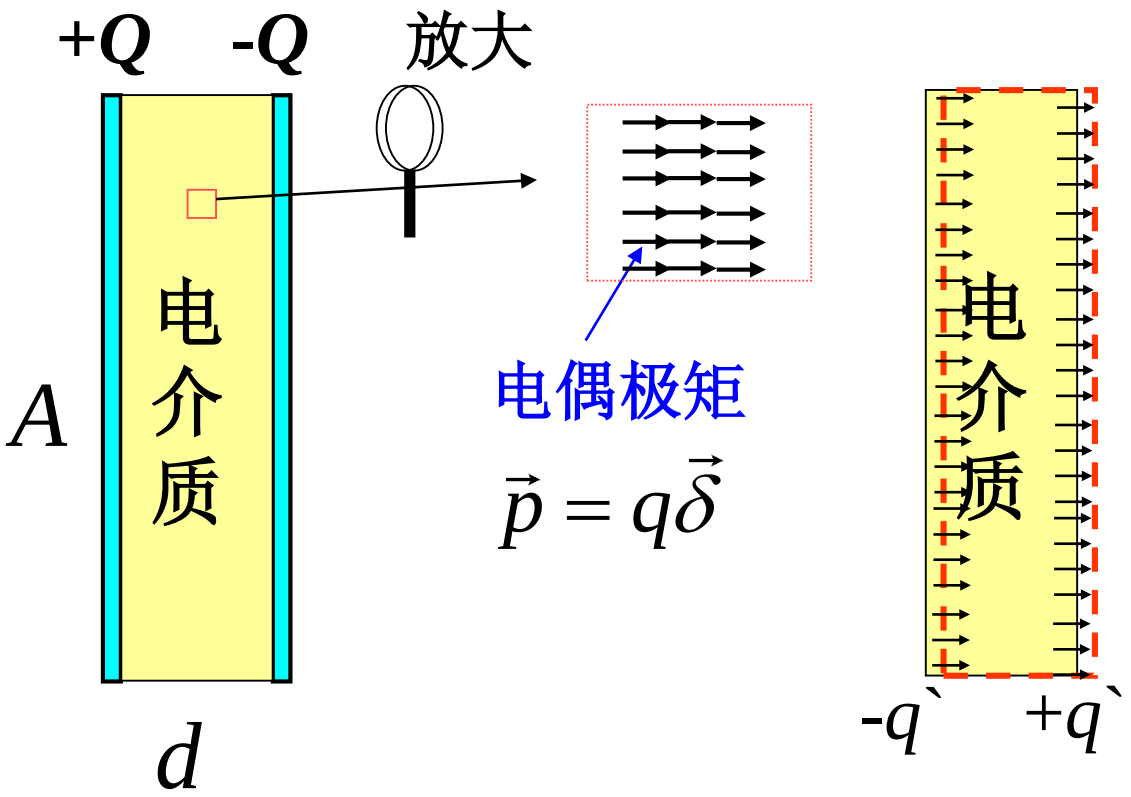


分子正负电荷中心在电场作用下错位

$$\vec{p} = q\vec{\delta}$$ 分子电偶极矩

位移极化 **(displacement) electronic polarization**

二. 电介质分子对电场的影响



$$E < E_0$$

$$U < U_0$$

$$U = \frac{U_0}{\varepsilon_r}$$

称极化电荷　　或称束缚电荷

Polarization charges　bound charges

考虑束缚电荷, 就不再考虑电介质

单选题 1分

设置

电介质对静电场的影响, 实际是通过极化电荷体现出来?

A 是

B 否

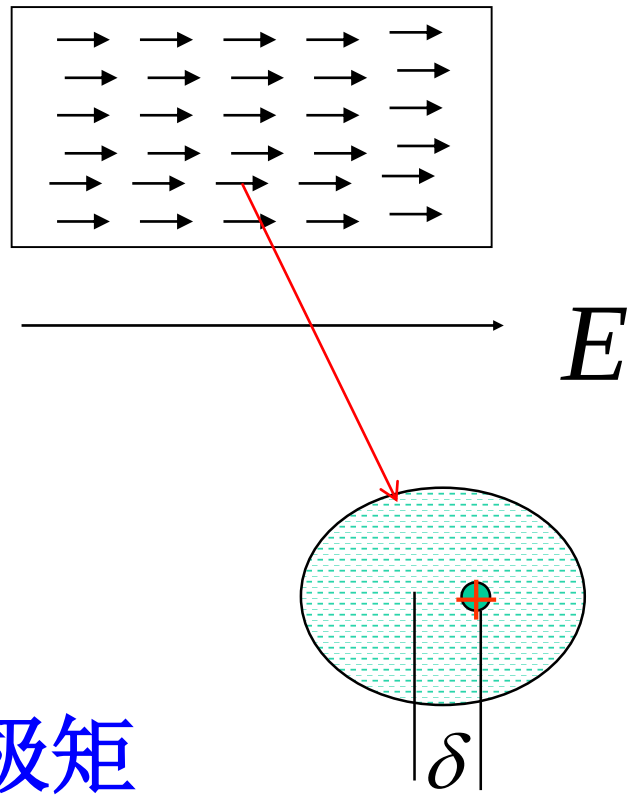
提交

三. 描述极化强弱的物理量--极化强度 $\vec{P}$

Polarization vector

定义:

单位体积的总电偶极矩



$$\vec{P} = N\vec{p}$$

分子数密度

单个分子位移极化的电偶极矩

平均电偶极矩

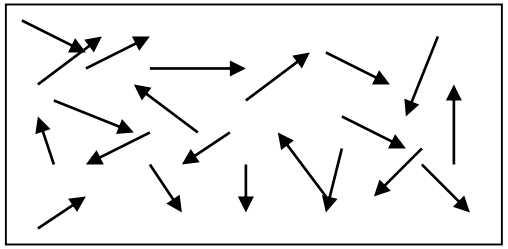
有极分子 polar molecules

$\vec{p}_{固有}$

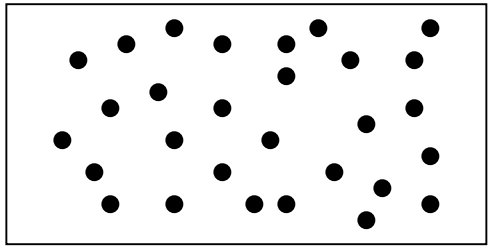
热运动 →无规取向

无外场时：

有极分子



无极分子



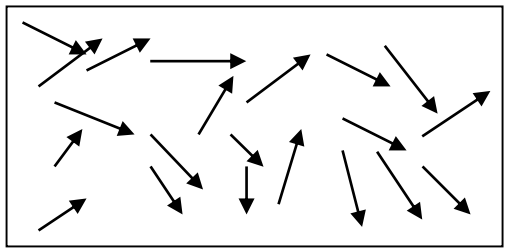
$\vec{E}$

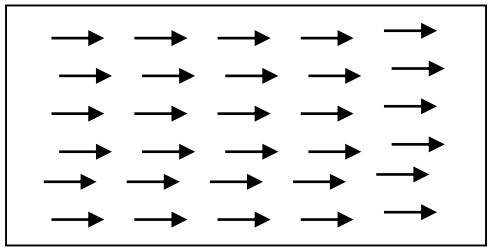
$\longrightarrow$

$-\vec{p}\cdot\vec{E}$

有外场时：

$$\boxed{\propto e^{\vec{p}\cdot\vec{E}/kT}}$$

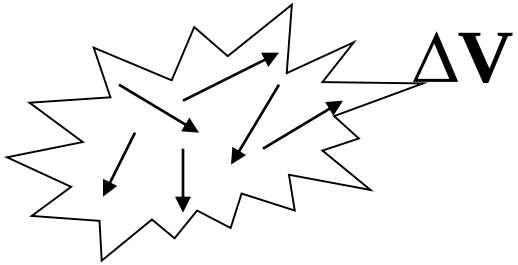




取向极化 (orientation polarization)

极化强度 $\vec{P}$　单位体积的总电偶极矩

$$\boxed{\vec{P} = \lim \frac{\sum_i \vec{p}_i}{\Delta V}}$$



宏观上无限小微观上无限大的体积元 ΔV

$\vec{p}_i$　每个分子的电偶极矩

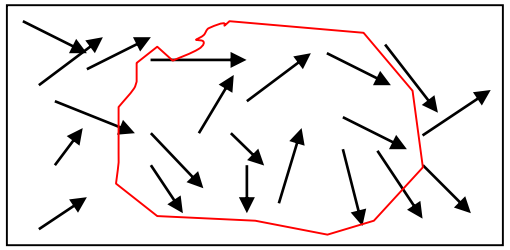
无外场时,无序排列, 求和为零　$\vec{P} = 0$

电偶极子排列的有序程度反映了介质被极化的程度

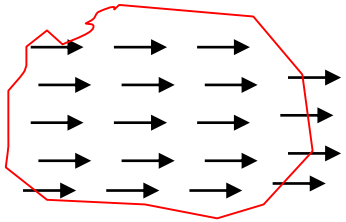
排列愈有序说明极化愈烈　$\vec{P}$ 越大　$\boxed{\vec{P}_{MAX} = N\vec{p}_{固有}}$

无极分子　上式 $= N\vec{p}$

有极分子 $\xrightarrow{\vec{E}}$



$\vec{p}_{固有}$



$$\vec{P} = \lim \frac{\sum_i \vec{p}_i}{\Delta V} = N\vec{p}$$

平均电偶极矩

考察小区域, 电场均匀

均匀极化

$\vec{P}$ 均匀

$$\boxed{\vec{p} = \frac{\vec{P}}{N}}$$

单选题 1分

设置

无论是有极分子组成的, 还是无极分子组成的电介质, 都可以在静电场中极化, 它们极化的机理是相同的?

A 是

B 否

提交

单选题 1分

设置

无论是有极分子极化, 还是无极分子极化, 都可以等效地用平均电偶极矩表示极化分子?

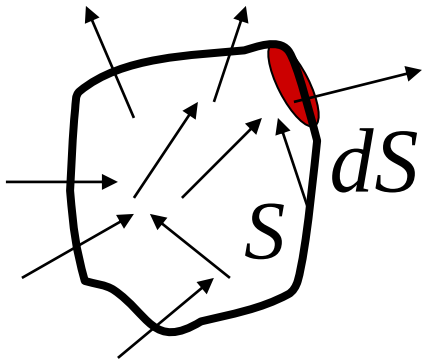
A 是

B 否

提交

四. 极化强度 $\vec{P}$ 与极化电荷的关系

在已极化的介质内任意作一闭合面S



S 内有多少极化电荷?

平均电偶极矩穿过S 的分子
对S内的极化电荷有贡献

平均电偶
极矩代表
一个分子

小面元dS贡献求和

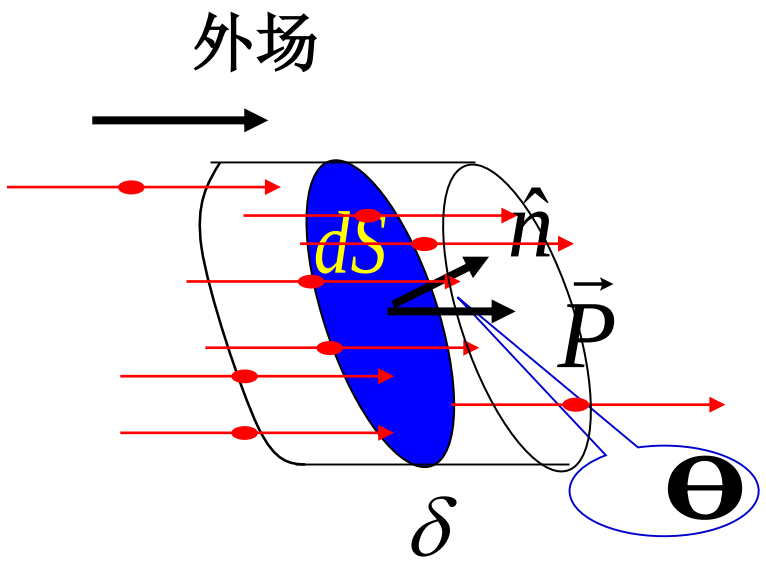
小面元*dS* 对面*S*内极化电荷的贡献

在*dS* 附近薄层内认为介质均匀极化

只考虑穿过*dS* 的分子或平均电偶极矩

平均电偶极矩

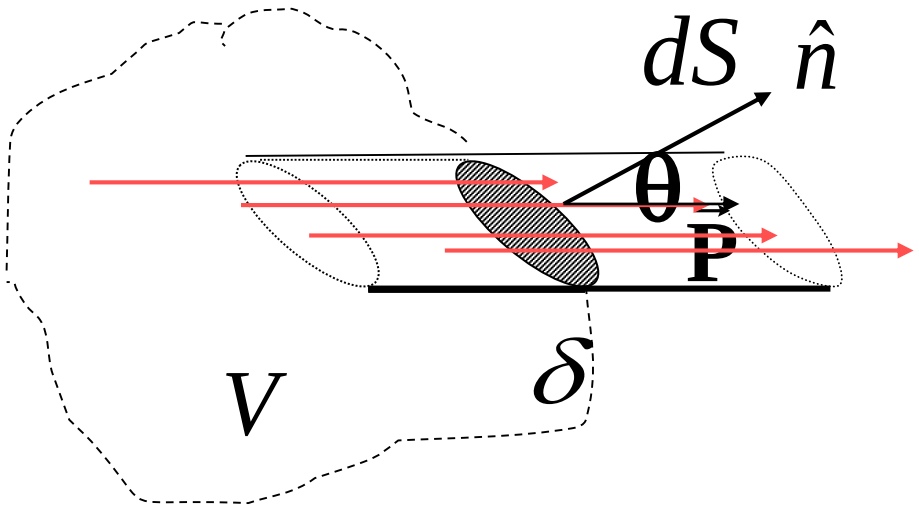
$$\vec{p} = q\vec{\delta} = \frac{\vec{P}}{N}$$



分子数密度N

$$|dq'| = |qN\delta\, dS\cos\theta| = |PdS\cos\theta| = |\vec{P}\cdot\hat{n}dS|$$

$$\left|dq'\right| = \left|\vec{P}\cdot\hat{n}dS\right|$$



小面元***dS***对
面内极化电荷的贡献

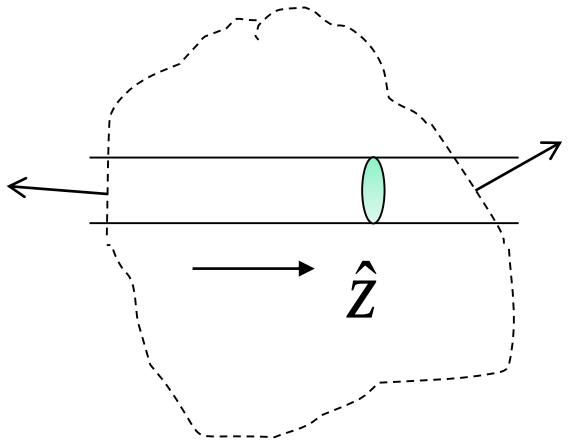
$$\boxed{dq' = -\vec{P}\cdot\hat{n}dS}$$

在***S***所围的体积内的极化电荷 q'与 $\vec{P}$ 的关系

$$\boxed{q' = -\oiint_S \vec{P}\cdot\hat{n}dS}$$

- 均匀极化处, 没有束缚电荷

$$q' = -\oiint_S \vec{P}\cdot\hat{n}dS = -P\oiint_S \hat{z}\cdot\hat{n}dS \quad = 0$$



- 均匀极化时束缚电荷只在表面
- 非均匀极化时束缚电荷可同时分布在内部和表面

单选题 1分

设置

均匀极化时, 电介质内部不出现极化电荷?

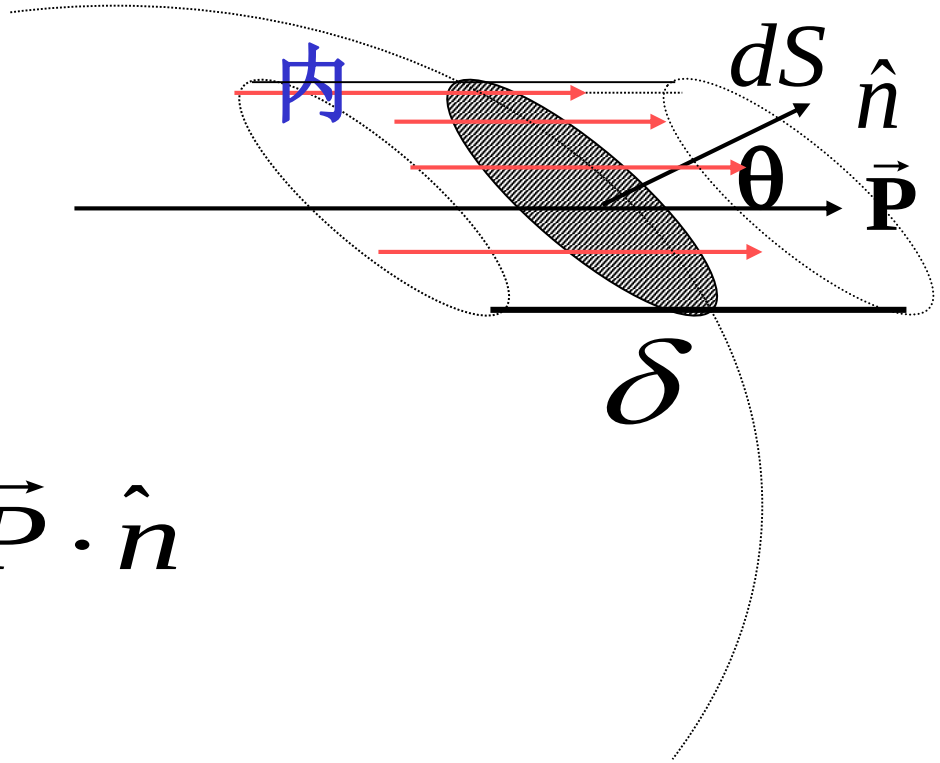
A 是

B 错

提交

电介质表面极化电荷面密度

$$dq' = \vec{P} \cdot \hat{n} dS$$



$$\sigma' = \frac{dq'}{dS} = \vec{P} \cdot \hat{n}$$

$$\boxed{\sigma' = \vec{P} \cdot \hat{n}}$$

$\hat{n}$ 介质外法线方向

五. 电介质的极化规律

1.各向同性线性电介质 isotropy linearity

$$\boxed{\vec{P} = \chi_e \varepsilon_0 \vec{E}} \quad \boxed{\chi_e = \varepsilon_r - 1}$$

介质的电极化率

χ_e 无量纲的纯数　　与 $\vec{E}$ 无关

2.各向异性线性电介质 anisotropy

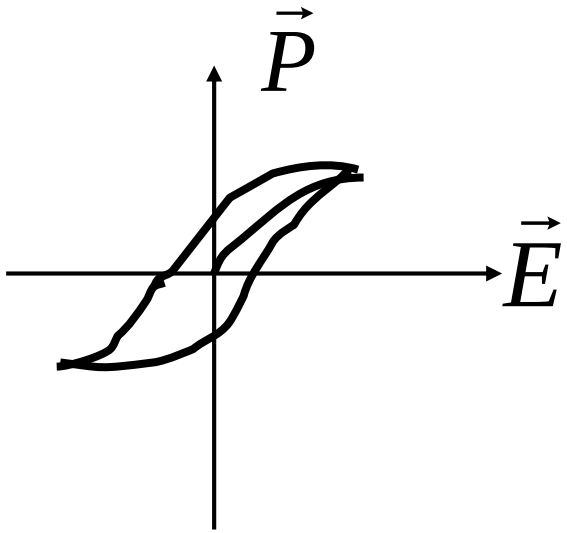
χ_e 与 $\vec{E}$ 、与晶轴的方位有关

张量描述　　弱场性质

3. 铁电体　　ferroelectrics

$\vec{P}$ 与 $\vec{E}$间非线性，

没有单值关系。



类似于铁磁体

主要宏观性质

1) 电滞现象

2) 居里点

3) 介电常数很大　ε_r　$10^2 \cdots 10^4$

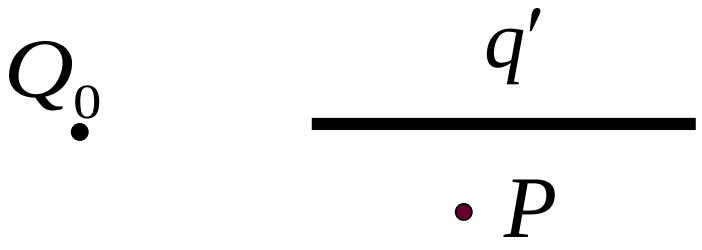
超声换能器

六. 有电介质时静电场的计算

自由电荷和极化电荷产生电场的叠加

$$\boxed{\vec{E} = \vec{E}_0 + \vec{E}'}$$

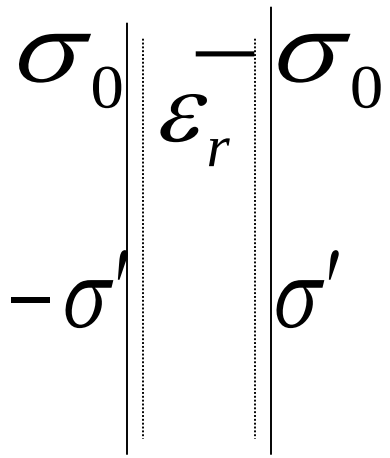
例1 介质细棒的一端放置一点电荷



P点的场强？

自由电荷和极化电荷共同产生

例2 平行板电容器　自由电荷面密度为 σ_0 充满相对介电常数为ε_r 的均匀各向同性线性电介质．求:板内的场



解:均匀极化

表面出现束缚电荷

所有电荷场叠加

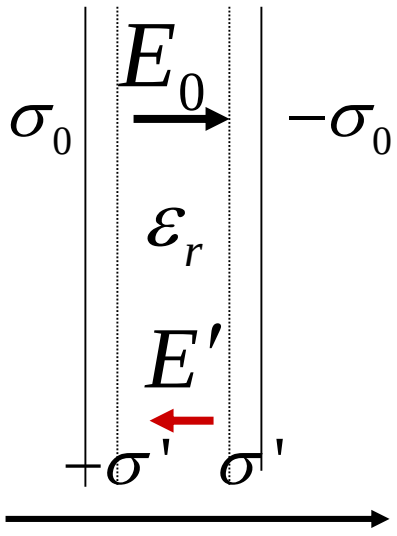
$$\pm\sigma_0 \longrightarrow E_0 = \frac{\sigma_0}{\varepsilon_0}$$

单独

$$\pm\sigma' \longrightarrow E' = \frac{\sigma'}{\varepsilon_0}$$

$$E = E_0 - E' = \frac{\sigma_0}{\varepsilon_0} - \frac{\sigma'}{\varepsilon_o}$$

共同产生



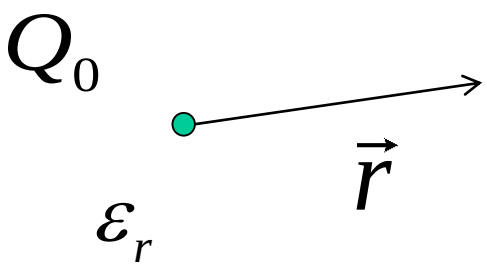
$$\sigma' = P_n = \varepsilon_0 \left(\varepsilon_r - 1\right) E$$

联立 →

$$\boxed{E = \frac{\sigma_0}{\varepsilon_0 \varepsilon_r}} \quad \boxed{= \frac{E_0}{\varepsilon_r}}$$

普遍?

例3 点电荷周围充满电介质时的电场



$$\vec{E} = \frac{Q_0}{4\pi\varepsilon_0\varepsilon_r} \frac{\vec{r}}{r^3}$$

§3 电位移矢量 $\vec{D}$（有电介质时的高斯定理）

electric displacement vector

一. 电位移矢量

$$\boxed{\vec{D} = \varepsilon_0 \vec{E} + \vec{P}}$$

量纲 $[\vec{D}] = [\vec{P}] = [\sigma]$ 单位 C/m

各向同性线性介质

$$\boxed{\vec{P} = \varepsilon_0 (\varepsilon_r - 1) \vec{E}}$$

介质方程

$$\boxed{\vec{D} = \varepsilon_0 \varepsilon_r \vec{E}}$$

$$\varepsilon = \varepsilon_0 \varepsilon_r$$

二. $\vec{D}$的高斯定理

$$\oiint_S \vec{E}\cdot d\vec{S} = \frac{\sum_i q_i}{\varepsilon_0} = \frac{\sum_i q_i' + \sum_i q_{oi}}{\varepsilon_0}$$

$$-\oiint_S \vec{P}\cdot d\vec{S}$$

$$\boxed{\vec{D} = \varepsilon_0 \vec{E} + \vec{P}}$$

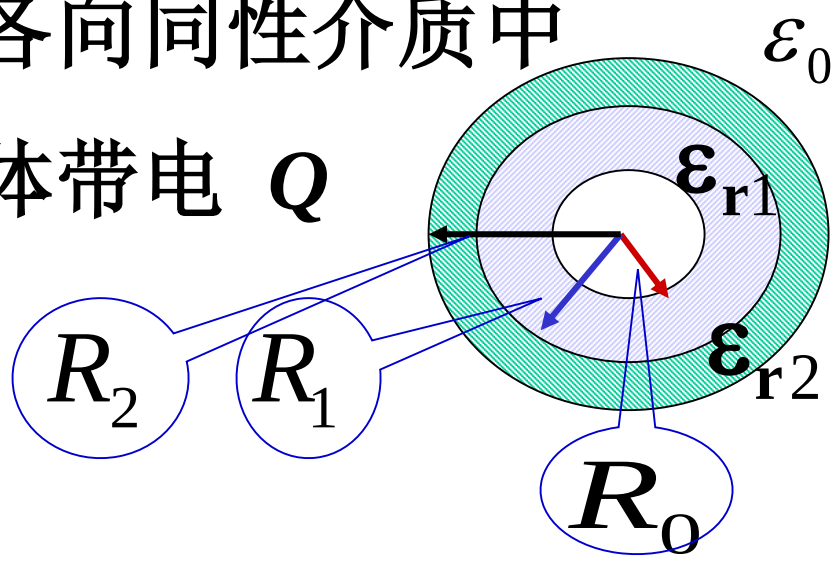
$$\boxed{\oiint_S \vec{D}\cdot d\vec{S} = \sum_i q_{0i}}$$

自由电荷

在具有某种对称性的情况下，可以首先解出 $\vec{D}$

即 $\vec{D} \Rightarrow \vec{E} \Rightarrow \vec{P} \Rightarrow \sigma' \Rightarrow q'$

例4　导体球置于均匀各向同性介质中（同心）如图示，导体带电 Q



求：

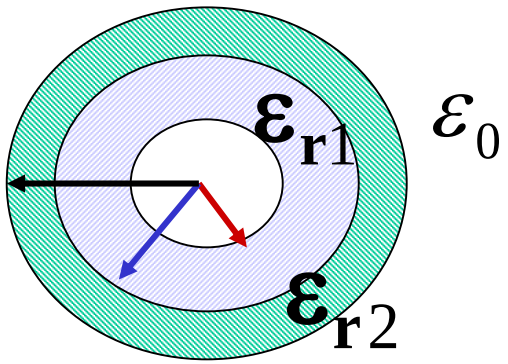
- 场的分布
- 紧贴导体球表面处的极化电荷
- 两介质交界处的极化电荷

解：

$$\oiint_S \vec{D}\cdot d\vec{S} = \sum_i q_{0i} \qquad \vec{D} = \begin{cases} 0 & r < R_0 \\ \dfrac{Q}{4\pi r^2}\hat{r} & r > R_0 \end{cases}$$

1) 场的分布

$$\vec{D}=\begin{cases}0 & r<R_0\\ \dfrac{Q}{4\pi r^2}\hat{r} & r>R_0\end{cases}$$



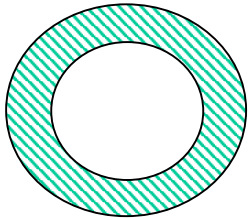
$$\boxed{\vec{E}=\frac{\vec{D}}{\varepsilon_0\varepsilon_r}}$$

$$\boxed{\vec{P}=\left(1-\frac{1}{\varepsilon_r}\right)\vec{D}}$$

按区域分别写出来

- 球对称极化, 不是均匀极化, 极化电荷体密度?

$$-q'=\oiint_S \vec{P}\cdot d\vec{S} = \text{常量}$$

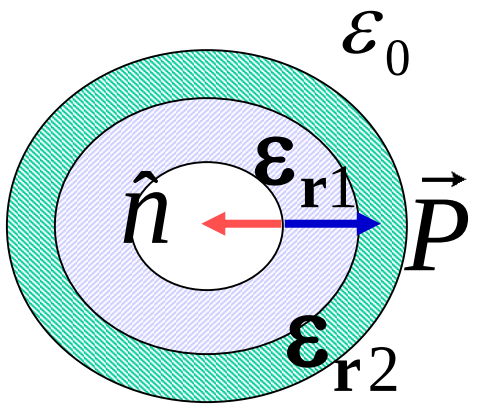


$$\rho'=0$$

2) 求紧贴导体球表面处的极化电荷

$$\sigma' = \vec{P}\cdot\hat{n}\Big|_{r=R_0} = -P_1\big|_{r=R_0}$$

$$\boxed{= -\frac{Q}{4\pi R_0^2}\left(1-\frac{1}{\varepsilon_{r1}}\right)}$$



$$q' = \sigma'\cdot 4\pi R_0^2 \boxed{= \left(\frac{1}{\varepsilon_{r1}}-1\right)Q}$$

思考: 点电荷 Q 周围的极化电荷?

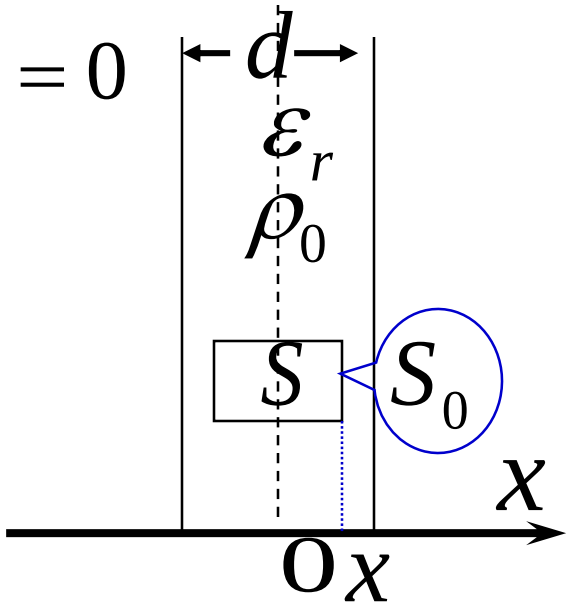
3) 两介质交界处极化电荷 $\sigma'\big|_{r=R_1} = (P_1 - P_2)\big|_{r=R_1}$

例5 一无限大各向同性均匀介质平板厚度为 d 相对介电常数为 ε_r 内部均匀分布体电荷密度为 ρ_0 的自由电荷

求：介质板内、外的 D E P

解： 面对称 $\vec{D}$ $\vec{E}$ $\vec{P}\perp$平板

取坐标系如图 $x=0$ 处 $E=0$



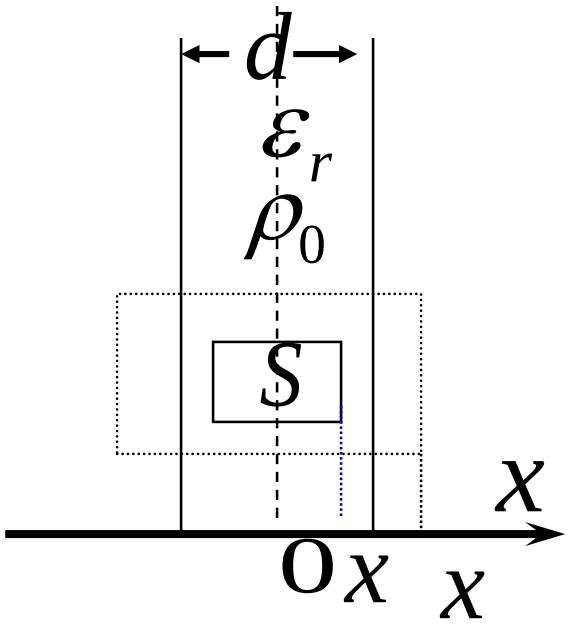
以 $x=0$ 处的面为对称

过场点作正柱形高斯面 S

底面积设 S_0

$$|x| \le \frac{d}{2} \qquad 2DS_0 = \rho_0 2|x| S_0$$

$$\boxed{D = \rho_0 |x|}$$



$$|x| \ge \frac{d}{2} \qquad 2DS_0 = \rho_0 S_0 d$$

$$\boxed{D = \frac{\rho_0}{2} d}$$

$$\vec{P} = \left(1 - \frac{1}{\varepsilon_r}\right)\vec{D}$$

极化体电荷密度?

$$\rho' = -\left(1 - \frac{1}{\varepsilon_r}\right)\rho_0$$

单选题 1分

设置

如果各向同性线性介质是均匀的，也就是介质内各处介电常数都相同，此时，介质内极化电荷密度与自由电荷密度成正比?

A 是

B 否

提交

$$|x| \le \frac{d}{2} \qquad \boxed{D = \rho_0 |x|} \qquad E = \frac{D}{\varepsilon_0 \varepsilon_r} = \frac{\rho_0 |x|}{\varepsilon_0 \varepsilon_r}$$

极化电荷面密度?

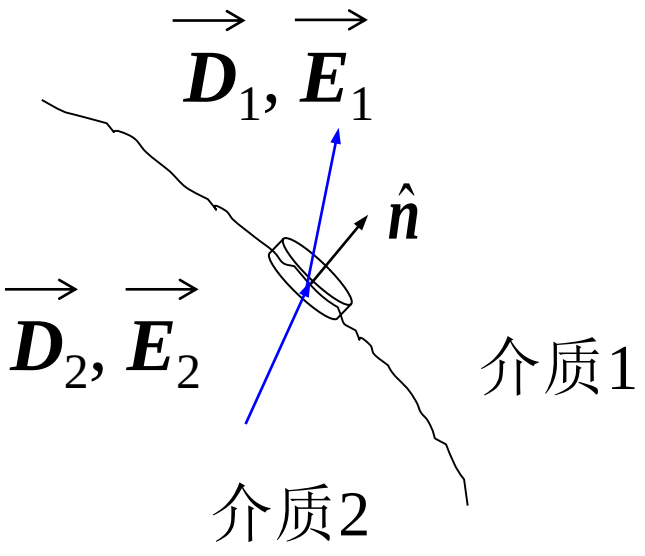
$$P = \left(\varepsilon_r - 1\right)\frac{\rho_0 |x|}{\varepsilon_r}$$

$$|x| \ge \frac{d}{2} \qquad \boxed{D = \frac{\rho_0}{2} d} \qquad E = \frac{D}{\varepsilon_0} = \frac{\rho_0 d}{2\varepsilon_0}$$

均匀场

$$P = \varepsilon_0 \left(\varepsilon_r - 1\right) E = 0$$

三. 边值关系



$$\oiint_S \vec{D}\cdot \mathrm{d}\vec{S} = \iint_{\text{底面1}} \vec{D}\cdot \mathrm{d}\vec{S} + \iint_{\text{底面2}} \vec{D}\cdot \mathrm{d}\vec{S}$$

$$= \vec{D}_1 \cdot \hat{n}_1 \Delta S + \vec{D}_2 \cdot \hat{n}_2 \Delta S$$

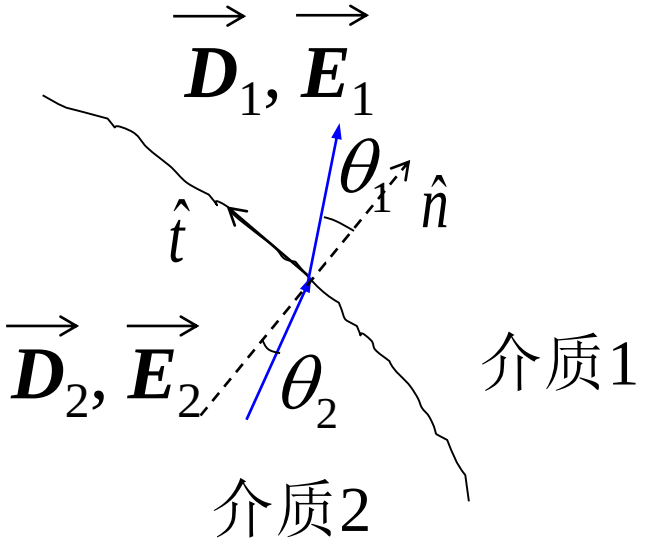
$$= (D_{1n} - D_{2n})\Delta S$$

$$(D_{1n} - D_{2n})\Delta S = q_0$$

如果分界面上
没有自由电荷

$$\boxed{D_{1n} = D_{2n}}$$

$$\boxed{D_{1n} - D_{2n} = \sigma_0}$$



$$\oint_L \vec{E} \cdot \mathrm{d}\vec{l} = (E_{1t} - E_{2t})\Delta l = 0$$

$$\boxed{E_{1t} = E_{2t}}$$

如果分界面上没有自由电荷

$$D_{1n} = D_{2n} \quad \rightarrow \quad \varepsilon_1 E_{1n} = \varepsilon_2 E_{2n}$$

$$\tan\theta = \frac{E_t}{E_n}$$

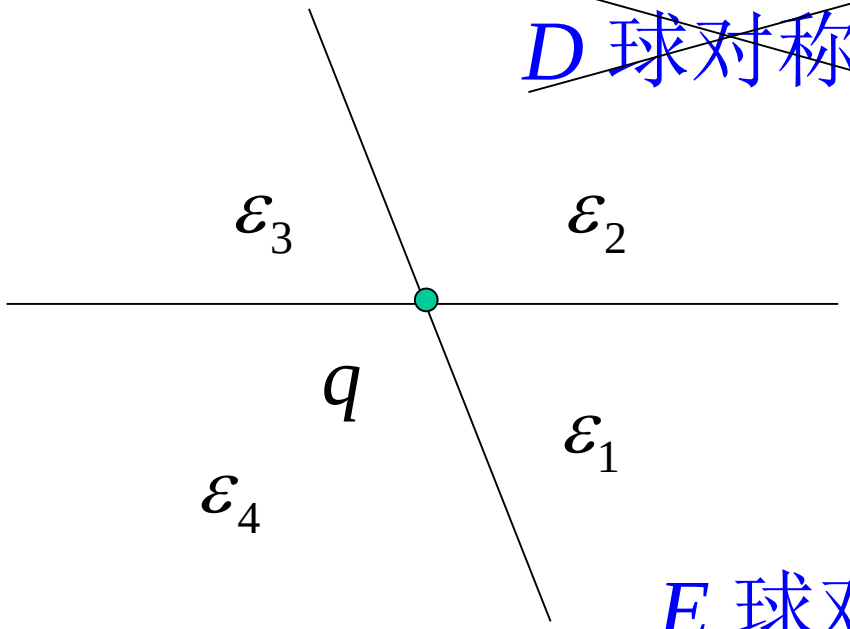
又 $E_{1t} = E_{2t}$

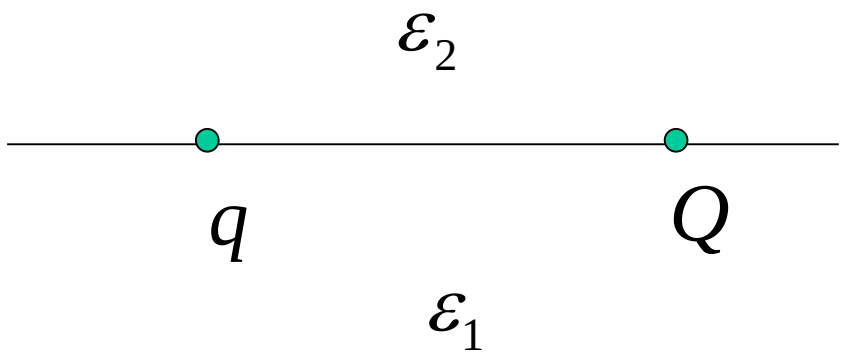
$$\frac{\tan\theta_1}{\tan\theta_2} = \frac{E_{1t}E_{2n}}{E_{1n}E_{2t}} = \frac{E_{2n}}{E_{1n}} = \frac{\varepsilon_1}{\varepsilon_2}$$

电位移线
或电场的
折射定理

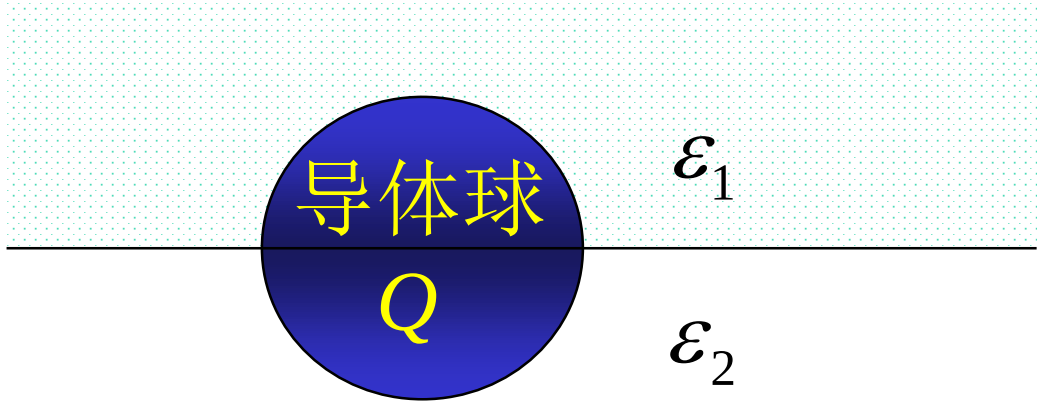
如下图, 电场为何?

~~D 球对称?~~



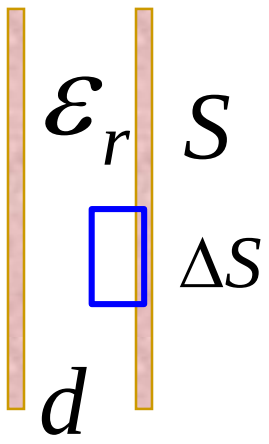


E 球对称!



§4 有电介质时静电场的能量

一. 有介质时的电容器的电容



$$\boxed{\varepsilon_r = \frac{C}{C_0}}$$

电容率

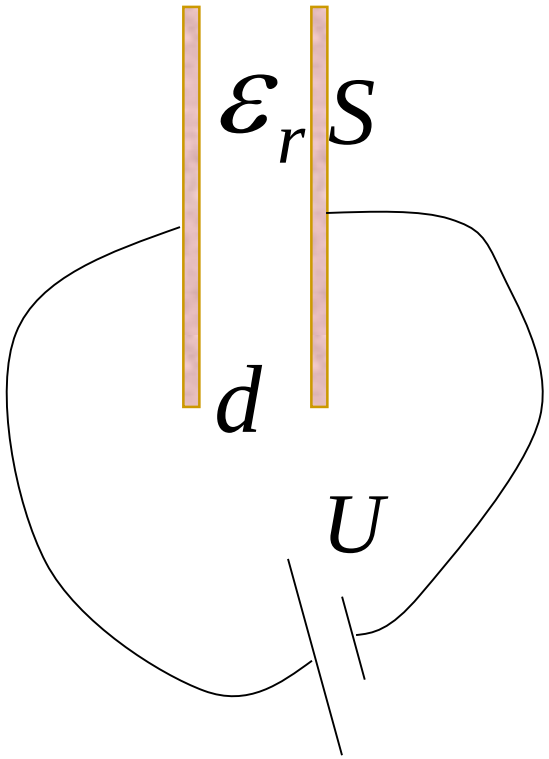
设带电 Q

$$C = \frac{Q}{U} \quad \rightarrow U = \frac{U_0}{\varepsilon_r} \quad \rightarrow E = \frac{E_0}{\varepsilon_r}$$

$$D = \varepsilon_0 \varepsilon_r E = \varepsilon_0 E_0 = \sigma_0 = \frac{Q}{S}$$

导体内 $D = 0$

二. 场能密度



静电场电势能比较复杂

充电　　电池做功 = 电容器储能

$$W_e = A = \int \underbrace{I}_{\frac{dq}{dt}} U dt = \int U dq$$

$$W_e = \int_0^Q \frac{q}{C} dq = \frac{Q^2}{2C} = \frac{1}{2} C U^2 = \frac{1}{2} Q U$$

有无介质 电容器电能公式相同

单选题 1分

设置

此题未设置答案，请点击右侧设置按钮

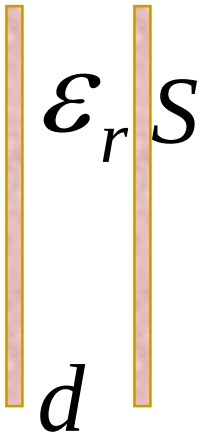
给定电压下, 给电容器充电. 假设有两个电容器, 一个充有电介质, 一个没有, 其它条件都完全相同, 这两个电容器充电后存储的能量相同.

A 是

B 否

提交

能量储存于场中



$$W_e = \frac{1}{2}QU = \frac{1}{2}DSEd = \frac{1}{2}DEV$$

电场能量密度为 $w_e = \dfrac{W_e}{V}$

$$\Longrightarrow \quad \boxed{w_e = \frac{1}{2}\vec{D}\cdot\vec{E}}$$

普遍